

$$\Leftrightarrow x^2 + 15x = 3250 \Leftrightarrow x^2 + 15x - 3250 = 0$$

$$\Leftrightarrow x^2 + 65x - 50x - 3250 = 0$$

$$\Leftrightarrow x(x + 65) - 50(x + 65) = 0$$

$$\Leftrightarrow (x + 65)(x - 50) = 0 \Leftrightarrow x = 50.$$

Hence, original number of persons = 50.

Ex. 49. Village X has a population of 68000, which is decreasing at the rate of 1200 per year. Village Y has a population of 42000, which is increasing at the rate of 800 per year. In how many years will the population of the two villages be equal?

Sol. Let the population of villages X and Y be equal after p years.

$$\text{Then, } 68000 - 1200p = 42000 + 800p \Rightarrow 2000p = 26000 \Rightarrow p = 13.$$

So, their population will be equal after 13 years.

Ex. 50. From a group of boys and girls, 15 girls leave. There are then left 2 boys for each girl. After this, 45 boys leave. There are then 5 girls for each boy. Find the number of girls in the beginning.

Sol. Let at present there be x boys. Then, number of girls at present = $5x$.

Before the boys had left: Number of boys = $x + 45$ and number of girls = $5x$.

$$\therefore x + 45 = 2 \times 5x \Leftrightarrow 9x = 45 \Leftrightarrow x = 5.$$

Hence, number of girls in the beginning = $5x + 15 = 25 + 15 = 40$.

Ex. 51. If x is an integer such that $x + \frac{1}{x} = \frac{17}{4}$, find the value of $x - \frac{1}{x}$.

(M.B.A., 2008)

$$\text{Sol. } x + \frac{1}{x} = \frac{17}{4} \Leftrightarrow \left(x + \frac{1}{x}\right)^2 = \left(\frac{17}{4}\right)^2 \Leftrightarrow x^2 + \frac{1}{x^2} + 2 \cdot x \cdot \frac{1}{x} = \frac{289}{16}$$

$$\Leftrightarrow x^2 + \frac{1}{x^2} = \frac{289}{16} - 2 = \frac{257}{16} \Leftrightarrow x^2 + \frac{1}{x^2} - 2 \cdot x \cdot \frac{1}{x} = \frac{257}{16} - 2$$

$$\Leftrightarrow \left(x - \frac{1}{x}\right)^2 = \frac{225}{16} = \left(\frac{15}{4}\right)^2 \Leftrightarrow x - \frac{1}{x} = \frac{15}{4}.$$

Ex. 52. A zoo keeper counted the heads of the animals in a zoo and found it to be 80. When he counted the legs of the animals he found it to be 260. If the zoo had either pigeons or horses, how many horses were there in the zoo?

(R.R.B., 2009)

Sol. Let the number of pigeons be x and the number of horses be y .

Then, total number of heads = $x + y$.

total number of legs = $2x + 4y$.

$$\therefore x + y = 80 \quad \dots(i)$$

$$\text{And, } 2x + 4y = 260 \text{ or } x + 2y = 130 \quad \dots(ii)$$

Subtracting (i) from (ii), we get: $y = 50$.

Putting $y = 50$ in (i), we get: $x = 30$.

Hence, number of horses in the zoo = 50.

Ex. 53. In a caravan, in addition to 50 hens there are 45 goats and 8 camels with some keepers. If the total number of feet be 224 more than the number of heads, find the number of keepers.

Sol. Let the number of keepers be x . Then,

Total number of heads = $(50 + 45 + 8 + x) = (103 + x)$.

Total number of feet = $(45 + 8) \times 4 + (50 + x) \times 2 = (312 + 2x)$.

$$\therefore (312 + 2x) - (103 + x) = 224 \Leftrightarrow x = 15.$$

Hence, number of keepers = 15.

Ex. 54. When an amount was distributed among 14 boys, each of them got ₹ 80 more than the amount received by each boy when the same amount is distributed equally among 18 boys. What was the amount?

Sol. Let the total amount be ₹ x . Then,

$$\frac{x}{14} - \frac{x}{18} = 80 \Leftrightarrow \frac{2x}{126} = 80 \Leftrightarrow \frac{x}{63} = 80 \Leftrightarrow x = 63 \times 80 = 5040.$$

Hence, total amount = ₹ 5040.